

Light Maturity Models (LMM): an Agile application

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ABSTRACT

‘Agile’ is become one of most known and used ICT buzzwords of last 10 years. More recently a plenty of ‘Agile Maturity Models’ were proposed, trying to evaluate the maturity (and capability) levels of an organization using such approach. The aim of this paper is to propose a feasible way to apply the ‘maturity model’ concept in a ‘light’ way for Agile applications and/or organizations, moving from the founding principle and values of the ‘Agile Manifesto’, in order to create more value from appraisals within a continual improvement program.

Categories and Subject Descriptors

D.2.9 [Management]: Software Process Models; K.6.3 [Software Management]: Software Process

General Terms

Management, Measurement, Verification.

Keywords

Agile, Maturity Models, Light Maturity Models, Crosby, Continual Process Improvement.

1. INTRODUCTION

‘Agile’ has become a buzzword in the ICT arena during last 15 years and it seems there is still a large confusion about its definition and contents. Is it a lifecycle (lightweight vs. heavyweight [4])? Is it an approach (‘old’ [2] and ‘new’ proposals for the Agile Manifesto [14][1][13][11])? Is it related to a development or project management style (ASD vs APM)? Is ‘Agile’ different or not from ‘Lean’ [21][15][38]? Or is it a mix of all of the above issues? Due to the large series of discussions and debates during last years [23], recently also the latest version of a well-known maturity model (MM) on software development such as CMMI-DEV [19] introduced some explanatory notes about agile practices in few Process Areas (CM, PI, PMC, PP, PPQA, RD, REQM, RSKM, TS and VER). But on the other side, many agilists prefer ‘Agile’ would be something separated and ‘pure’ from those models, mainly based on the ‘Agile Manifesto’ [2].

Thus the main question arising was: how to evaluate the level of adoption and maturity of such practices and values inside an organization/team? Maturity Models (MMs) can be a proper solution, improving the way a (Quality) Management System is appraised in the ISO world [26]. Thus, a plenty of proposals for an AMM (Agile Maturity Model) come out in the second part of last decade, but often poorly defined and with few details about how to appraise, compared with well-known appraisal processes such as those defined in SCAMPI [39] and ISO/IEC 15504-5 [29]. The less the details for a practical, continual process improvement, the less the business advantage when adopting them. But looking at the concrete, quantitative number of

enterprises regularly adopting some ‘maturity models’ (MMs) worldwide – whatever the application domains (see e.g. [20][41] for a first list of possible MMs) – they seem to be not so high because thought to be too costly and heavy to be run in particular in Small-Medium Enterprises (SMEs) and Very Small Entities (VSEs) [16], also at the light of the current economic global situation. Thus, a different solution must be put in action for achieving anyway improvements and better business results. And since Agile methods were initially thought and mostly applied to small teams and organizations, a similar solution could be adopted as yet done for SMEs/VSEs. Thus, a ‘light maturity model’ (LMM) – that would be a grid-based evolutionary model such as the one proposed 30 years ago by Crosby – but using one (or more) Agile principles and values as drivers for improvement, could be a valid alternative for that kind (and size) of organizations.

The paper is organized as follows: Section 2 presents the related works on some proposals for an “Agile Maturity Model”. Section 3 introduces the mechanics for a ‘light maturity model’ (LMM), moving from the original Crosby’s idea. Section 4 presents the possible ways to apply such generic concept in the Agile world. Finally, Section 5 draws some conclusions and discusses prospects for future works.

2. RELATED WORKS

Table 1 summarizes some of the specific proposals for an Agile MM, quite often labeled ‘AMM’:

Table 1. Maturity & Capability Models (MCM) on Agile

Model	Year	ML	Entity of Interest	Repr.	Level of Depth
Agile Maturity Model (AMM) [9]	2003	5 (0-4)	Org./Project	Staged	Low
Agile Maturity Model [36]	2006	7 (-1/5)	Org./Project (5 dimensions)	Staged	Low
Agile Maturity Model [24]	2008	7	Project (9 dimensions)	Staged	Medium
Agile Maturity Model [37]	2008	6 (-1/3+)	Org./Project (10 dimensions)	Staged	Low
Agile Scaling Model (ASM) [6][5] (initially aka as APMM - Agile Process MM)	2009	3	Org./Project (8 scaling factors)	Staged	Medium
Agile Maturity Model (AMM) [7]	2010	5 (1-5)	Org./Project	Staged	Low

It can be observed that there is not uniformity in the number of maturity levels (ML), in the number and types of

dimensions/scaling factors to be analyzed. The only common element is the ‘staged’ representation (not the ‘continuous’ one) and a low level of description of the elements to be rated and the eventual ‘best practices’ to be followed for each dimension/scaling factor to achieve higher capability and maturity levels.

3. LIGHT MATURITY MODELS (LMM)

Moving from the global picture depicted in previous sections, the business question to be posed would mainly be: “how could we achieve better business results adopting agile methods and techniques and measure their level of adoption?” and “which could be an evolutionary path for a continual (agile) process improvement initiative?”. Some initial requirements & constraints (RC) from the SME/VSE/Agile viewpoint:

- **RC1:** Appraisal(s) needed *but* should be cheap, quick and sufficiently detailed for writing effective improvement plans;
- **RC2:** Appraisal method(s) should be simply to be understood and produce short reports for management, showing in 1 slide what is going on and where to intervene for improving results;
- **RC3:** A clear definition of most relevant drivers for improvement should be done and shared across the organization/team;
- **RC4:** A series of ‘best practices’ for each of the selected drivers must be chosen as basis for describing the ‘roadmap’ to be followed [17][30].

For coping with **RC1-RC2**, Fig. 1 shows the current and shared manner to appraise by MMs using the *continuous* representation.

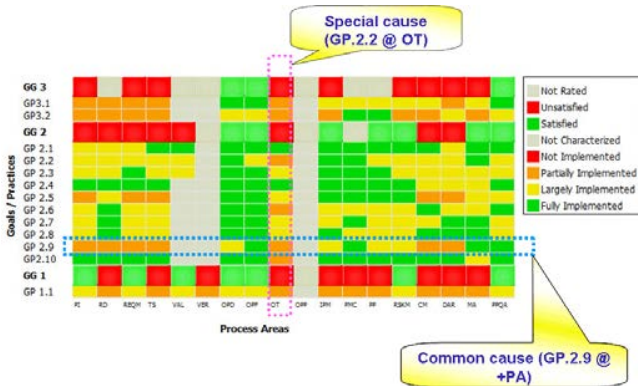


Figure 1. A typical continuous appraisal with CMMI (special vs. common causes)¹

It allows – by a rating on a 4-level ordinal scale (N/P/L/F – Not/Partially/Largely/Fully achieved) – to assign to each appraisal driver (e.g. GP – Generic Practice – in the CMMI language; PA – Process Attributes in ISO 15504, etc.) to describe a profile by each appraised process (by column) and by each single driver (by row). The result from analysis, as show, can easily stress the possible presence of common/special causes, providing tips for an effective and specific list of actions in the improvement plan.

¹ Original picture kept from the SQI Appraisal Assistant website (<http://www.sqi.gu.edu.au/AppraisalAssistant/>)

With respect to **RC1** (to be cheap, quick and sufficiently detailed), ISO come back to roots with Appendix A of the 9004:2009 standard [28]. Such Appendix returned on the usage of the initial Crosby’s Quality Management Maturity Grid [33][34], proposing 5 evolutionary *stages* (columns) and a certain number of *appraisal drivers* (rows). Each crossing cell by each row contains a brief sentence describing the status for that driver at that level. Knowing which is the achieved level and the next one for each driver of interest will quickly allow to write down an improvement plan with specific actions. ISO 9004 – that’s a complementary norm to ISO 9001[27] – proposed therefore as *drivers* the requirements contained into the norm (from §4 on) and provided a yet ‘template’ ready for use and overcoming the ISO 19011 way to audit a Management System (e.g. for ISO 9001, 20000-1, 27001, 50001, etc).

But the question at this stage is: how to rate if we achieved (or not) a certain level? Do we have to use a binary scale (on/off; yes/no) or a more detailed one? Thus, the logical answer, leading to a general schema we call ‘Light Maturity Models’ (LMM), is the adoption of the typical 4-level ordinal scale (N/P/L/F) adopted by most known models (CMMI; SPICE), even if using a different % distribution across levels. Figure 2 is an elaboration from an ISO 9004 snapshot, proposing the % rating scale from ISO 15504-5.



Figure 2. ISO 9004:2009 Appendix A: coming back to roots (our elaboration on [28])

The advantage against the solely adoption of a grid-based approach as the Crosby’s one besides in looking the whole picture and possibly rating – as in MMs – each driver. Furthermore, knowing the causal relationships among the different *drivers* (in the case of ISO 9001, the *clauses* of the standard, representing the answer to **RC3**) and their positioning in the reference process model – whatever it is - it will be easier to determine the priority for two or more possible actions to run according to what is the deeper causes for a certain effect to be removed.

Again, the adoption of a LMM approach represents also a sort of *bridge* between the typical ‘audit’ language and approach in the ISO language and the appraisal process in most advanced MMs. In such way, people learn to use a more granular schema for evaluating their way of working (possibly more and more) by objective evidences (OE) per each requirement (and associated processes). In fact, using mappings between the *drivers* and the process reference model (PRM) adopted (e.g. CMMI), it could be possible to yet have high-level indications about the typical work products/artifacts to be expected for a certain process/goal, that’s not suggested by the solely application of list of ‘requirements’ to be accomplished with such as in ISO 9001, 20000-1 etc., simply

because they specify the ‘*what*’ is needed to be done and not the ‘*how*’ to do it. This logical link by a mapping table allows to cope with **RC4**.

Some attention points to follow:

- Define the *drivers* (dimensions/scaling factors) in a clear, non-ambiguous and measurable way: one of the main problems when evaluating an entity of interest (EoI) is to do share the same exact definition among different people. So, if it’s true that ‘*you cannot control what you cannot measure*’, coming back ‘*you cannot measure what you cannot define*’. From a well-communicated and non-ambiguous definition, it will be possible – as also Crosby did – to insert some possible quantitative measures for each cell of the grid, in order to help appraisers to choose and rate the proper capability level². Moving from a clear definition of EoIs, it will be possible to apply e.g. a GQM (Goal-Question-Metric)-like approach as a technique for deriving the proper measures from the needed information goals.
- Link the *drivers* to 1+ preferred PRM practices: often these drivers are requirements (as in the ISO management systems standards) or principles/values (as in the ‘Manifesto’), not processes their selves, as in the typical, full MMs as CMMI or ISO 15504. Thus a *bridge* must be created. An external mapping table from the short word/concept to the related process(es) in the preferred PRM(s) for the organization could be a quick and feasible work product to create and share across all the stakeholders;
- Create an evolutionary, harmonic path toward higher capability levels by each driver: it’s fundamental to validate a proper balancing about the statements describing two consequent levels. The risk is to ask too (or too few) for a certain level and too few (or too) for the next one, increasing the difficulty to properly catching a real picture of your organization/project at that time.

A first adoption of such LMM approach was the **EM³ base model** [31] for an Energy Management System, based on the new upcoming ISO 50001 standard [18][25].

4. LMMs FOR AGILE

Next step is to contextualize the *general-purpose* structure of a whatever LMM to Agile. Since the few proposal for an ‘Agile Maturity Model’ listed in Section 2, Table 2 lists the dimensions/scaling factors they proposed, that could be used as the title of the *rows* of your Agile LMM.

Table 2. AMM drivers for an Agile LMM

AMM [36]	AMM [24]	AMM [37]	ASM/APMM [6][5]
Org./Project (5 dimensions) ³	Project (9 dimensions) ⁴	Org./Project (10)	Org./Project (8 scaling)

² From a Process Improvement viewpoint, ‘capability level’ is about a process/tasks, while ‘maturity level’ as an overall rating for the whole organization/project, even if in current language ‘maturity’ would refer to a series of evolutionary stages.

³ These are the definitions for the 5 dimensions: **Testing** - the extent to which testing is integrated into day-to-day development activities; **Collective code ownership** - how aggressively people pair and can commit code anywhere in a

		dimensions) ⁵	factors) ⁶
Testing	Testing	Testing	Team Size
Collective code ownership	Source Code Mgmt	Requirements	Geographical Distrib.
Collaboration	Collective code ownership	Build	Regulatory Compliance
Assurance/Gov ernance	Collaboration	Shared Responsibility	Domain Complexity
Simplicity	Responsiveness to Business	Governance	Org. Distribution
	Assurance/Governance	Communication	Tech. Complexity
	Story Formation	Config. Mgmt	Org. Complexity
	Design Simplicity	Simplicity	Enterprise Discipline
	Build Process	Assurance	
		Responsiveness	

Another possibility is to move directly from the 12 principles in the Agile Manifesto [2], that could be reinforced also by the ‘SPI Manifesto’ [3], sharing and deploying more in detail several principles and values yet expressed by the Agile Manifesto.

Note that the above presented are example applications that need to be filled and completed according to the peculiarities of your own organization/project. But the idea is to start smoothly with such kind of approach more than continuing with the solely application of a ‘yes/no’ logic or with the solely criteria proposed in ISO 19011 standard, also if the interested organization is a SME/VSE. Improvement can be scaled according to the organizational size and constraints, but needs to be more quantitative as possible. And starting with a shared definition of the issue under evaluation represents the first step in order to achieve it and be more and more able during time in clearly detects the *deeper cause* for a certain effect. Adopting an evolutionary approach such these ones depicts yet a possible path allowing an agile organization/team to quickly and easily improve but starting to look also at the organizational/support processes and not focusing strictly on the primary processes, that’s typical of an agile team (and ASD methodology).

Last but not least, note that such kind of appraisal run into an organization/team yet using an APM method such as SCRUM can reinforce that project management style and it’s not a sort of double effort: a *burndown chart* reports the ‘time’ dimension as

system; **Collaboration** - the events and frequency with which developers, management and customer interact and communicate; **Assurance/governance** - the degree to which management activities integrate directly into development tasks; **Simplicity** - the extent to which developers are locked-in to a technical design or have freedom to refactor with each new requirement.

⁴ This AMM yet proposed a label for each of the foreseen 7 evolutionary stages by each driver.

⁵ This is a refinement of [36]; this presentation does not include definitions for such drivers.

⁶ ASM/APMM focus more on stressing the opposites from an organizational viewpoint (simple vs. complex).

the overall summary of an agile planning, but cannot itself provide all the needed information for re-planning the project.

Therefore a LMM for your preferred AMM could be a cheap, quick and sufficiently detailed complementary solution to the yet applied APM method for improving business results.

5. CONCLUSIONS & NEXT STEPS

Continual improvement [42] is a goal for any organization, no matter its size, business domain, and application context. Agile methodologies represent a possible tool for emphasizing the way SME/VSE and – when properly adopted – also large organizations can achieve better technical and business results. But such improvement must be measurable by a series of evolutionary steps towards higher capability levels for each of the main improvement drivers useful for that organization.

Light Maturity Models (LMM) can be considered a bridge between the non-usage and the full usage of MMs, hoping to provide also for SME/VSE a cheaper way to overcome the typical efficacy/effectiveness criteria from the ‘audit’ world while moving towards measurable ‘appraisals’. And a plenty of information can arise from process appraisals, creating value when properly addressed with specific corrective/improvement actions.

The few MMs for Agile formulated right now at a high-level shows that this field undergoes a rapid change and is subjected to reflection and evaluation by both practitioners and researchers. It is logical to expect even further elaboration of the agile maturity concept in the near future. The combination of an LMM logic with the preferred Agile model/approach could therefore reinforce the project management practices for those organizations/teams pursuing to be more and more ‘agile’ during time. And the further value from process appraisal results asking a very few effort can be kept also by a small team for its project and not necessarily only at the ‘organizational unit’ (OU) level, as typically done with audits and appraisals.

Our future work efforts therefore will be devoted in proposing some case studies using one of those possible applications of Agile LMM.

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